Examiner only

Answer **all** questions.

1. (a) The table below contains statements about the motion of an object and the forces acting on it.
For each statement, place **one** tick (✓) to show which law it applies to.
One row has been completed as an example. [3]

Statement	Newton's 1st Law	Newton's 2nd Law	Newton's 3rd Law
A skydiver accelerates when the forces are unbalanced.	✓		
This law can be written as $F = ma$.			
A train will not move from rest unless acted on by a resultant force.			
When it is fired, a rifle exerts a force on a bullet. The bullet exerts an equal and opposite force on the rifle.			

(b) A ball is dropped from rest (u = 0 m/s) from a window.
It takes a time, t = 1.2 s to reach the ground.

(i) Use the equation:

$$v = u + at$$

to calculate the speed, v, of the ball as it hits the ground. [2]
(Acceleration, a = 10 m/s^2)

v = ... m/s

(ii) Use your answer above for v and the equation:

$$x = \frac{u + v}{2}\, t$$

to calculate the distance, x, of the window above the ground. [2]

x = ... m

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